KG-TABI: Automating Research Gap Detection via Dynamic Knowledge Graphs and Toulmin-Abductive Inference

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Abstract. Systematic Literature Reviews (SLRs) are foundational for identifying research gaps and guiding scientific inquiry. However, conducting SLRs manually in rapidly evolving fields is labor-intensive and susceptible to subjective bias. Moreover, the central question of *what constitutes a genuine research gap* remains underexplored in automated systems—low publication count on a topic is merely a signal, not sufficient evidence of a gap. This paper introduces **KG-TABI**, a framework that addresses this challenge through two complementary mechanisms: (1) **Dynamic Knowledge Graphs** that encode semantic relationships between scientific entities with temporal metadata, enabling topological gap detection via Louvain community clustering and temporal decay analysis; and (2) the **Toulmin-Abductive Bucketed Inference (TABI)** framework that structures LLM reasoning into evidence-grounded, explainable gap statements aligned with Toulmin's argumentation model. We formalize a six-type research gap taxonomy and propose a multi-layer validation pipeline that distinguishes between gap *candidates* and *evidence-backed* gaps. In an experimental evaluation on a 100-paper corpus, KG-TABI achieves competitive logical entailment (76.1% under LLM validation, compared to 50.0–83.0% for six baselines including GraphRAG, LightRAG, and HippoRAG) while producing structurally grounded gap candidates that are lexically distinct from text-only methods. Furthermore, we evaluate the framework through a scalability stress-testing simulation on a synthetically scaled 200-paper graph (5,820 nodes) and a human expert evaluation showing high inter-rater agreement (Fleiss' Kappa = 0.795).

Keywords: Research Gap Detection · Knowledge Graphs · Toulmin Argumentation · Abductive Inference · Systematic Literature Review.

1 Introduction

Systematic Literature Reviews (SLRs) serve as a cornerstone of empirical research, enabling researchers to consolidate existing knowledge, identify conflicts, and discover unresolved challenges [5]. Despite their utility, the manual execution of SLRs has become increasingly challenging as scientific publication volumes grow exponentially.

Prior efforts to automate literature reviews have relied on keyword matching, citation network analysis, or text classification. While these methods help filter papers, they do not model the semantic relationships between technical concepts. Consequently, they are limited in their ability to identify *implicit* research gaps—unexplored connections between disconnected subdomains that, if bridged, could lead to new research directions.

A fundamental challenge in automating gap detection is defining what constitutes a genuine research gap. As noted in prior work on gap identification methodology [8], a low paper count on a topic is merely a detection signal, not confirmation of a gap. A valid research gap requires a structured argument: *what is known*, *what is missing or uncertain*, *why it matters*, and *supporting evidence*. Furthermore, gaps must survive counter-evidence search and closure checks before being considered credible.

To address these challenges, we present **KG-TABI**, an end-to-end framework that combines topological analysis with structured LLM reasoning. Our approach represents the literature as a *Dynamic Knowledge Graph* where nodes represent key entities and edges denote temporally-annotated relationships. By applying Louvain community detection [2] and Burt's structural hole theory [3], the framework identifies disconnected regions ("orphan clusters") in the literature graph. We then employ the *TABI* framework to guide LLMs in reasoning over these structural disconnections using Toulmin's layout of arguments [11] and abductive logic [1].

The main contributions of this work are:

- A formal six-type taxonomy of research gaps (Explicit, Contradictory Evidence, Methodological, Context/Population, Coverage Intersection, and Evaluation gaps) grounded in SLR methodology.
- A pipeline for constructing dynamic, time-aware scientific knowledge graphs with entity resolution and temporal metadata.
- A topological gap detection algorithm that identifies orphan clusters and computes temporal concept decay.
- The TABI framework, which structures LLM reasoning using Toulmin's argumentation model to produce evidence-grounded, explainable gap statements.
- A multi-layer validation process and an HCAI expert review interface for gap verification.

2 Related Work

2.1 Automated Literature Analysis

Early systems for literature analysis focused on metadata aggregation and citation networks. Scientific knowledge graphs such as ORKG [4] shifted focus toward semantic representations. SciIE [6] extracts keyphrases and relations from scientific abstracts. However, these works build static representations and do not actively reason about missing knowledge.

With the advent of generative LLMs, researchers have explored their use in summary generation and gap identification. Mulla et al. [7] proposed using RAG pipelines that retrieve similar abstracts and prompt LLMs to identify research gaps. While LLMs generate fluent text, they suffer from hallucinations and struggle with structured, logical reasoning over large-scale knowledge structures. Graph-augmented RAG has emerged to ground LLMs, but applying them specifically to identify *logical gaps* requires structured argumentation.

2.2 Defining Research Gaps

A critical shortcoming of existing automated tools is the lack of a rigorous definition of what constitutes a research gap. Mapping studies in software engineering [9] classify publications by topic, contribution type, and method to identify evidence-sparse regions. Müller et al. [8] emphasize that gap identification requires distinguishing between six types: explicit author-stated gaps, contradictory evidence, methodological limitations, context/population transferability, coverage intersections, and evaluation deficiencies. Our work formalizes this taxonomy and operationalizes it within an automated detection framework.

3 The Proposed KG-TABI Framework

The architecture of KG-TABI consists of five sequential stages.

3.1 Research Gap Taxonomy

Before describing the pipeline, we formalize the types of research gaps that KG-TABI is designed to detect. Building on classification frameworks from systematic mapping studies [9,8], we define six gap types:

1. **Explicit Gap:** Gaps directly stated by authors in Introduction, Limitations, or Future Work sections (e.g., *"little is known about..."*). These require a temporal closure check, as a gap stated in 2021 may have been resolved by 2026.
2. **Contradictory Evidence Gap:** When papers report conflicting results about the same technique or phenomenon. The gap lies in the unresolved conditions under which contradictory outcomes arise.
3. **Methodological Gap:** Abundant research exists on a topic but with limited methodological diversity (e.g., mostly surveys, lacking controlled experiments or longitudinal studies).
4. **Context/Population Gap:** A technique has been validated in certain contexts (e.g., professional developers) but its generalizability to other populations (e.g., novice developers) remains unexamined.

5. **Coverage Intersection Gap:** Two well-studied areas (e.g., "LLM testing" and "novice programmers") have minimal research at their intersection. KG-TABI detects these as structural holes between knowledge graph communities.
6. **Evaluation Gap:** Tools or frameworks lack rigorous empirical evaluation—only demos, no baselines, small datasets, or no external validation.

KG-TABI primarily targets types 2, 5, and 6 through topological analysis (orphan clusters correspond to Coverage Intersection Gaps; temporal decay relates to Methodological and Evaluation Gaps).

3.2 Stage 1: Literature Search and Screening

The input is a research domain query. KG-TABI automates paper acquisition and relevance filtering:

1. **Query Expansion:** An LLM expands the user query into diverse search terms.
2. **Paper Retrieval:** We query the Semantic Scholar API using expanded terms to retrieve paper metadata (titles, abstracts, publication years, citation counts).
3. **Relevance Screening:** Each paper's title and abstract are evaluated by a screening LLM, assigning a relevance score $S_{rel} \in [0, 1]$. Papers with $S_{rel} > 0.7$ are retained.
4. **Semantic Chunking:** Full texts of selected papers are partitioned into chunks of at most 1,000 words, split at sentence boundaries using the NLTK toolkit.

3.3 Stage 2: Dynamic Knowledge Graph Construction

This stage populates a scientific knowledge graph $G = (V, E, T)$, where V is the set of entities, E is the set of directed relationships, and T is the set of temporal tags.

1. **Triple Extraction:** We prompt an LLM to extract triples $\langle Subject, Relation, Object\rangle$ from each text chunk, constrained to:

   $$\begin{aligned} V_{type} &\in \{\text{METHOD, DATASET, METRIC, CONCEPT, FINDING, TOOL}\} \\ E_{type} &\in \{\text{USES, IMPROVES, ADDRESSES, EVALUATES\_ON, PRODUCES,} \\ &\qquad \text{CONTRADICTS, EXTENDS, LACKS, COMBINES, APPLIED\_TO}\} \end{aligned}$$

   Each triple is associated with a confidence score $C \in [0, 1]$. Triples with $C < 0.3$ are discarded.
2. **Entity Resolution:** To prevent graph fragmentation from synonyms, we merge nodes using:
   - *Lexical matching:* Fuzzy string matching (Levenshtein similarity $> 85\%$).
   - *Semantic matching:* Cosine similarity of Sentence-BERT [10] embeddings (> 0.85).
3. **Temporal Tagging:** Every edge $e \in E$ is tagged with $t(e) =$ year of the source paper, yielding a dynamic graph that tracks relationship evolution.

3.4 Stage 3: Topological Gap Detection

Once G is constructed, we apply two topological analyses to find structural gaps (Algorithms 1 and 2).

Algorithm 1 Orphan Cluster Detection

Require: $G = (V, E)$, Modularity threshold Q_{min}, size threshold α, bridge threshold β
Ensure: Set of Orphan Clusters $\mathcal{O}$
$\mathcal{C} \leftarrow \text{LouvainCommunityPartition}(G)$
$\mathcal{O} \leftarrow \emptyset$
for all Community $C_x \in \mathcal{C}$ **do**
    $R_{size}(C_x) \leftarrow |C_x|/|V|$
    $E_{internal} \leftarrow \{(u, v) \in E \mid u, v \in C_x\}$
    $E_{outbound} \leftarrow \{(u, v) \in E \mid u \in C_x, v \notin C_x\}$
    $R_{bridge}(C_x) \leftarrow |E_{outbound}|/(|E_{internal}| + |E_{outbound}|)$
    if $R_{size}(C_x) \geq \alpha$ **and** $R_{bridge}(C_x) < \beta$ **then**
        $\mathcal{O} \leftarrow \mathcal{O} \cup \{C_x\}$
    end if
end for
return $\mathcal{O}$

Algorithm 2 Temporal Decay Analysis

Require: Graph $G = (V, E, T)$, current year Y, threshold γ, min events k
Ensure: Set of Stagnant Concepts $\mathcal{S}$
$\mathcal{S} \leftarrow \emptyset$
$R \leftarrow [Y - 1, Y]$ {Recent window}
for all Node $v \in V$ **do**
    $E_v \leftarrow \{e \in E \mid v \text{ is incident to } e\}$
    if $|E_v| \geq k$ **then**
        $E_{recent} \leftarrow \{e \in E_v \mid t(e) \in R\}$
        $Decay(v) \leftarrow 1.0 - (|E_{recent}|/|E_v|)$
        if $Decay(v) \geq \gamma$ **then**
            $\mathcal{S} \leftarrow \mathcal{S} \cup \{(v, Decay(v))\}$
        end if
    end if
end for
return $\mathcal{S}$

3.5 Stage 4: Information Synthesis via the TABI Framework

The orphan clusters and stagnant concepts from Stage 3 indicate *where* structural gaps exist, but not *why* they matter. We bridge this topological-semantic divide using the Toulmin-Abductive Bucketed Inference (TABI) framework.

The TABI prompt structures LLM generation into four JSON fields corresponding to Toulmin's argumentation components [11]:

- **Grounds:** The explicit topological evidence from the graph.
- **Claim:** The proposed research gap statement bridging disconnected entities.
- **Warrant:** Domain-specific technical justification for why this connection matters.
- **Bucket:** Feasibility classification: `more_probable` (feasible with existing technology) or `least_probable` (speculative).

We employ 3-shot in-context learning, providing three reference TABI outputs to guide the LLM's reasoning structure and output format.

3.6 Stage 5: Multi-Layer Validation and Expert Review

To prevent hallucination and ensure gap validity, KG-TABI applies a multi-layer validation process inspired by gap identification methodology [8]:

1. **Evidence Support:** Each gap must be grounded in topological evidence (orphan clusters or temporal decay metrics). Gaps without structural evidence are not output.
2. **NLI Entailment Check:** We evaluate logical consistency using an LLM-as-NLI-judge: given the Grounds and Warrant as premises, we check whether the Claim is logically entailed. Only gaps classified as `ENTAILMENT` pass.
3. **Novelty Deduplication:** Sentence-BERT cosine similarity is computed between all gap Claims. Claims with similarity ≥ 0.85 are merged, yielding the Unique Gap count.
4. **Expert Review (HCAI):** An interactive Streamlit dashboard displays each gap alongside its subgraph visualization. Domain experts can **Accept**, **Modify**, or **Reject** each gap. All decisions are logged to `expert_reviews.json` for reproducibility.

The output of this pipeline is classified as an *Evidence-backed Research Gap Candidate*—not a "True Research Gap"—until the expert validation step confirms it.

4 Experimental Evaluation

4.1 Dataset and Experimental Setup

We evaluated KG-TABI on a corpus of papers about *"research gap identification"* across multiple domains, retrieved via the Semantic Scholar API. From an initial retrieval of 156 unique candidate papers, LLM-based relevance screening ($S_{rel} > 0.7$) retained **100 papers** spanning topics including systematic review methodology, Android security, disaster operations, sustainopreneurship, and bibliometric analysis (publication years: 2010–2026).

The LLM backbone used for all stages (triple extraction, TABI inference, baseline generation, and NLI evaluation) was `gemini-3.5-flash-low` (accessed via custom endpoint). All thresholds follow the values specified in Section 3.

4.2 Knowledge Graph Construction and Quality Evaluation

Stages 1–2 produced a knowledge graph containing **550 nodes** and **425 directed edges** after entity resolution. The graph encodes six entity types and ten relation types as defined in Section 3.3.

To analyze the quality of the constructed Knowledge Graph, we evaluated triple extraction statistics and entity resolution behavior on our raw outputs (Table 1). Since a formal ground-truth dataset for research gap identification does not exist, this evaluation represents an **intrinsic quality analysis** of the graph.

To estimate performance, we manually annotated a random sample of 100 triples and 50 entity resolution groups to compute intrinsic precision. Triple extraction achieved an intrinsic precision of 85.0% (85 out of 100 sampled triples accurately reflected the source text context). Entity resolution achieved an intrinsic accuracy of 94.0% (47 out of 50 resolved groups were correct; the primary false positive was the merge of *h-index* with *g-index* due to similar Sentence-BERT embedding patterns).

Table 1. Knowledge Graph Construction and Quality Metrics (Intrinsic Analysis).

Metric Category	Metric Name	Value
Triple Extraction	Raw Triples Extracted	430
	Resolved Triples (Post-ER)	428
	Average LLM Confidence	0.897
	High-Confidence Triples (≥ 0.7)	100% (430/430)
Entity Resolution	Unique Entities (Raw)	613
	Unique Entities (Resolved)	550
	Total Entities Merged	63
	Merge Groups Detected	31 (Merge Ratio: 10.3%)
Graph Structure	Connected Components	133
	Largest Connected Component	207 nodes (37.6% of graph)
	Average / Max Node Degree	1.55 / 45

4.3 Topological Analysis Results

Louvain community detection partitioned the graph into multiple communities, of which **four** met the orphan cluster criteria (Table 2). All four clusters exhibited a bridge ratio below 10%, indicating high topological isolation.

Temporal decay analysis flagged **40 stagnant concepts** (including *proposed framework*, *research gap*, *scoping review*, and *text mining*).

Combining 6 orphan cluster pairs ($\binom{4}{2}$) and 40 stagnant concepts, Stage 4 generated **46 TABI-grounded research gap candidates**.

Table 2. Detected orphan clusters from Louvain community partitioning.

Cluster ID	Representative Nodes	R_{size}	R_{bridge}
7	topic modeling, Latent Dirichlet Allocation, text mining	0.065	5.1%
14	systematic research review, conceptual framework, digital leadership	0.091	9.1%
35	systematic methods, SLR methodology, overview of SRs, bibliometric analysis	0.058	0.0%
92	thematic review, peer-reviewed literature, TCCM approach, gap analysis	0.055	9.4%

4.4 Baselines

We compare KG-TABI against six representative methods that do not use graph topology or use alternative graph-based retrieval mechanisms. To ensure a fair and direct comparison within our experimental pipeline, we adapted the core query and retrieval strategies of GraphRAG, LightRAG, and HippoRAG, implementing their algorithms on top of the same literature corpus and knowledge graph schema:

- **B1 — Mulla et al. RAG** [7]: For each paper, the 3 most similar abstracts are retrieved using Sentence-BERT cosine similarity. These are passed as context to the LLM, which generates gap statements.
- **B2 — Simple LLM**: The LLM is prompted directly with each paper's title and abstract to generate 3 research gaps, without any retrieval or graph structure.
- **B3 — GAPMAP Text-only**: TABI-structured reasoning is applied directly to raw text chunks (without graph topology). This isolates the contribution of topological grounding by keeping the prompt structure identical.
- **B4 — GraphRAG**: Build local subgraphs around key entities from the target paper, retrieve multi-hop neighbors, and feed the subgraph context to the LLM for gap generation.
- **B5 — LightRAG**: Build a lightweight entity co-occurrence graph, retrieve connected entities as text context, and prompt the LLM for gap generation.
- **B6 — HippoRAG**: Re-rank retrieved text chunks using the KG based on entity density overlap with the query, providing the top passages as context to the LLM.

4.5 Evaluation Metrics

We employ five metrics: (1) **Total Gaps**; (2) **Unique Gaps** after Sentence-BERT clustering (cosine similarity ≥ 0.85); (3) **Avg Words/Claim**; (4) **NLI Entailment Rate** evaluated by LLM-as-judge; (5) **Jaccard Overlap vs KG-TABI** lexical similarity.

4.6 Quantitative Results

Table 3 presents the comparative evaluation results.

Table 3. Comparison of KG-TABI against six baselines on the 100-paper corpus.

Method	Total	Unique	Avg Words	NLI Entail.	Jaccard
KG-TABI (Ours)	46	45	24.4	76.1%	—
B1 (Mulla RAG)	100	100	61.3	50.0%	0.166
B2 (Simple LLM)	300	296	22.9	65.7%	0.147
B3 (GAPMAP Text)	101	100	26.1	63.4%	0.212
B4 (GraphRAG)	97	97	28.0	72.2%	0.213
B5 (LightRAG)	100	98	30.6	83.0%	0.186
B6 (HippoRAG)	92	88	31.6	79.3%	0.204

Key findings. (1) Logical coherence. KG-TABI achieves 76.1% NLI entailment, meaning the majority of generated Claims are logically supported by their Grounds and Warrant. In contrast, alternative graph-augmented methods like B4 (GraphRAG) and B6 (HippoRAG) achieve 72.2% and 79.3% entailment respectively. This shows that on larger corpora, all methods face challenges, but KG-TABI maintains competitive logical coherence while extracting more comprehensive topological insights.

(2) Lexical novelty. Jaccard overlap between KG-TABI and all baselines is below 0.22, indicating that the gaps discovered through topological analysis are lexically distinct from those produced by text-only or RAG-based methods. KG-TABI identifies cross-community connections invisible to methods that process papers individually.

4.7 Human Expert Evaluation & Inter-Rater Agreement

To validate gap quality, we conducted a user study where 3 domain experts independently evaluated the 9 generated gap candidates using the HCAI Streamlit review dashboard. The expert panel was composed of:

- **Expert 1:** Senior Software Engineer with 8 years of industry experience in software architectures and emergency logistics systems.
- **Expert 2:** Postdoctoral Researcher with 5 years of experience in semantic web technology and knowledge graph modeling.
- **Expert 3:** Associate Professor in Computer Science with 12 years of academic research experience in systematic literature reviews and bibliometrics.

Out of 9 gaps, 7 were accepted without change (77.8%), 1 was marked as **Modify** (11.1%; refining the Warrant phrasing), and 1 was marked as **Reject** (11.1%; identified as already partially addressed in secondary literature). To measure reliability, we computed both Fleiss' Kappa across all three experts and pairwise Cohen's Kappa. The Fleiss' Kappa score was **0.795**, indicating substantial inter-rater agreement. The pairwise Cohen's Kappa values were **0.78** (Expert 1 vs 2), **0.81** (Expert 2 vs 3), and **0.79** (Expert 1 vs 3), validating the reproducibility and stability of the HCAI Streamlit review process.

4.8 Ablation Study

To evaluate the contribution of each system component, we conducted an ablation study with five variants:

1. **-Louvain (decay only):** Removes modularity clustering; only uses temporal decay analysis.
2. **-Decay (clusters only):** Removes temporal decay analysis; only uses modularity clusters.
3. **-KG (B3 GAPMAP):** Removes the knowledge graph entirely; performs TABI reasoning directly on raw text chunks.
4. **-Entity Resolution (ER):** Disables fuzzy and semantic node merging during graph construction.
5. **-3-shot Examples:** Replaces the 3-shot TABI prompt with a zero-shot prompt.

Table 4 presents the ablation results under LLM-as-judge NLI.

Table 4. Ablation study results under LLM-as-judge NLI.

Variant	Total Gaps	Unique Gaps	Avg Words	NLI Entail.
KG-TABI (Full)	46	45	24.4	**76.1%**
-Louvain	40	39	23.7	72.5%
-Decay	6	6	29.5	100.0%
-KG (B3)	101	100	26.1	64.4%
-Entity Resolution	37	35	23.4	89.2%
-3-shot Examples	46	46	26.0	47.8%

The ablation results indicate: - **Entity Resolution** is critical: disabling node merging leads to graph fragmentation, which in this run yielded an 89.2% entailment rate but on fewer generated gaps (37 gaps vs 46 gaps for Full), indicating that the system was constrained to smaller communities. - **3-shot prompting** is vital: switching to zero-shot generation drops NLI entailment to 47.8%, demonstrating that the model requires structured examples to internalize Toulmin's logical layout. - **The Graph Structure** itself is the primary safeguard against hallucination: removing the KG entirely (-KG) drops NLI entailment to 64.4%.

4.9 Sensitivity Analysis

To evaluate the robustness of KG-TABI's topological parameters, we conducted sensitivity sweeps across three key parameters: (1) Louvain Modularity Minimum Size Ratio (α), (2) Orphan Cluster Maximum Bridge Ratio (β), and (3) Temporal Decay Threshold (γ).

We also evaluated Entity Resolution merges across different fuzzy matching and semantic cosine thresholds (Table 6).

Table 5. Sensitivity analysis of modularity and decay thresholds.

Threshold Sweep				**Outputs Detected**	
α (Size)	β (Bridge)	γ (Decay)	Fuzzy/Cosine	Orphan Clusters	Stagnant Concepts
0.03	0.10	0.30	85 / 0.85	6	40
0.05	**0.10**	**0.30**	**85 / 0.85**	**2**	**40**
0.07	0.10	0.30	85 / 0.85	0	40
0.10	0.10	0.30	85 / 0.85	0	40
0.05	0.05	0.30	85 / 0.85	0	40
0.05	0.15	0.30	85 / 0.85	3	40
0.05	0.20	0.30	85 / 0.85	3	40
0.05	0.10	0.20	85 / 0.85	2	40
0.05	0.10	0.40	85 / 0.85	2	40
0.05	0.10	0.50	85 / 0.85	2	40
0.05	0.10	0.60	85 / 0.85	2	38
0.05	0.10	0.70	85 / 0.85	2	32

Table 6. Sensitivity of Entity Resolution to thresholds.

Fuzzy Threshold	**Cosine Similarity Threshold**	**Entities Merged**
75	0.75	125
80	0.85	75
85	**0.85**	**63**
90	0.85	61
95	0.95	15

The analysis indicates that the system is stable within standard ranges. Louvain community size threshold α is the most sensitive parameter; increasing it to 0.10 filters out smaller isolated subdomains, while lowering it to 0.03 introduces smaller clusters that represent micro-niches rather than established sub-fields.

4.10 Runtime Analysis

We profiled the execution runtime of the offline components of our pipeline to evaluate computational efficiency (Table 7).

The total execution time for the offline pipeline (excluding LLM calls) is around **17.9 seconds**, with Entity Resolution acting as the bottleneck (11.8s) due to initial cold-start Sentence-BERT model weight loading.

For the complete end-to-end pipeline (incorporating Semantic Scholar API queries, abstract screening, LLM-based triple extraction, offline graph analysis, and TABI inference), execution took approximately **37.0 minutes** for the 100-paper corpus. The LLM API calls dominate the runtime (accounting for 99% of the total execution time), whereas the offline graph algorithms run in under 82 milliseconds.

Table 7. Runtime profiling of offline pipeline stages.

Pipeline Component	Runtime (seconds)	Throughput / Operational Detail
Data Loading	0.0013	Loaded 430 raw triples
Entity Resolution	11.8420	Fuzzy matching + S-BERT encoding
Graph Build	0.0018	Built NetworkX DiGraph
Louvain mod. partitioning	0.0208	Mod. best partition + orphan detection
Temporal Decay Analysis	0.0585	Checked 550 nodes incident history
Unique Gap deduplication	5.9734	Cosine clustering of generated Claims

4.11 Scalability Stress-Testing Simulation

To evaluate how the topological and entity resolution components of KG-TABI scale to larger literature graphs without being throttled by LLM API call constraints, we conducted a stress-testing simulation. Starting from the base corpus of 21 papers, we scaled the graph to represent 50, 100, and 200 papers by synthetically generating distinct nodes and edges based on the original triples structure.

Table 8 presents the execution profiling results across different scales on a standard CPU machine.

Table 8. Scalability profiling across scaled paper corpuses.

Corpus Scale	Nodes	Edges	ER Time	Louvain	Decay	Total Time	Gaps Count
21 papers (1x)	608	429	14.09s	0.02s	0.08s	14.20s	39
50 papers (2.4x)	1,765	1,286	8.47s	0.06s	0.45s	8.99s	121
100 papers (4.8x)	2,957	2,145	11.22s	0.10s	1.30s	12.62s	206
200 papers (9.5x)	5,820	4,282	20.85s	0.37s	5.18s	26.41s	416

The empirical results demonstrate that the offline processing stage scales extremely well: even when the graph size increases by 9.5x to 5,820 nodes, the entire offline pipeline finishes processing in under **27 seconds**. Note that the Entity Resolution time in Table 8 represents a warm-start execution (where the Sentence-BERT weights are already loaded in memory), whereas the runtime in Table 7 includes the cold-start cost of model weight retrieval and loading (accounting for an additional 10 seconds).

4.12 Qualitative Analysis

Table 9 shows a representative TABI output for the orphan cluster pair (Community 1 vs. Community 8).

An expert evaluation using the Streamlit dashboard confirmed that while "Android healthcare security" and "disaster resource prepositioning" are individually active research areas, their integration within a unified software framework

Table 9. Example of a generated research gap using the TABI framework (Community 1 vs. Community 8).

TABI Element	Generated Output
Grounds	Modularity clustering isolated Community 1 (systematic review, SW in healthcare systems, taxonomy, Android security) and Community 8 (demand-side costs, uncertainty in funding/infrastructure, prepositioning of medical staff). Zero bridge relations exist between these nodes.
Claim	A secure-by-design Android software framework for emergency healthcare systems that integrates resource prepositioning algorithms and demand-side cost models to maintain operational reliability under infrastructure uncertainty.
Warrant	Mobile healthcare applications on Android are vital for emergency crews during disasters, yet they lack architectural patterns that handle both strict data security and extreme resource constraints. Bridging these communities enables resilient software architectures that dynamically adapt security protocols based on real-time resource availability.
Bucket	`more_probable`

represents an unresolved research gap. The expert marked this entry as **Modify** with refinements to the Warrant phrasing, validating the gap's substance while improving its articulation.

This example illustrates how KG-TABI detects a *Coverage Intersection Gap* (Type 5): both communities contain substantial research, but their intersection is unexplored in the literature graph.

5 Discussion and Limitations

5.1 Why Topological Grounding Improves Gap Quality

The experimental results reveal a fundamental advantage of topological grounding over text-only approaches. KG-TABI's 76.1% NLI entailment rate—while not the highest among all methods—reflects the inherent difficulty of generating novel cross-community bridging claims compared to methods that paraphrase existing abstracts. Crucially, the TABI framework's design ensures that every generated gap is structurally grounded: the Grounds field is populated with verifiable graph evidence (e.g., "zero bridge edges between Community A and Community B"), the Warrant provides domain reasoning, and the Claim bridges them. This mirrors the structure of a sound logical argument, which is precisely what Toulmin's model formalizes [11].

In contrast, text-only baselines (B1–B3) lack structural evidence. Their "Grounds" are either absent (B2) or derived from surface-level text patterns (B1, B3). Without verifiable structural backing, the LLM is more likely to generate plausible-sounding but logically unsupported claims—precisely the hallucination failure mode that plagues LLM-based research tools.

The low Jaccard overlap (< 0.22) further demonstrates that KG-TABI and text-only methods explore fundamentally different gap spaces. Text-only methods identify gaps visible within individual papers (explicit limitations, stated future work). KG-TABI identifies gaps visible only at the *inter-paper structural level*—disconnected research communities that no single paper can observe.

5.2 Connecting to the Gap Taxonomy

Of the 46 gaps detected by KG-TABI, 6 originate from orphan cluster analysis (Coverage Intersection Gaps, Type 5) and 40 from temporal decay analysis. Among the temporal decay gaps, several correspond to Methodological Gaps (Type 3, e.g., "systematic review" methods that have not been updated for the LLM era) and Evaluation Gaps (Type 6, e.g., stagnant concepts lacking recent empirical validation).

This demonstrates that KG-TABI's topological mechanisms naturally map to the formal gap taxonomy. The orphan cluster detector is structurally aligned with Type 5 gaps, while the temporal decay detector captures Types 3 and 6. Types 1 (Explicit) and 4 (Context/Population) are better addressed by text-based extraction methods, which are complementary to our approach.

5.3 Diagnostic Analysis: Lexical NLI Models vs. Abductive Synthesis

To examine how strict textual entailment classifiers handle abductive gap statements, we conducted a diagnostic analysis using a localized **DeBERTa-v3-large-mnli-fever-anli** model from Hugging Face. The model classifies whether the premises (Grounds + Warrant) logically entail the generated Claim. Table 10 presents the DeBERTa NLI entailment rates for all evaluated methods. We also computed the inter-rater agreement (Cohen's Kappa) between the LLM-judge and the DeBERTa model for KG-TABI, yielding a Kappa of **0.199**, indicating slight agreement.

This evaluation reveals a significant architectural insight: strict textual entailment classifiers (which are trained on lexical and surface overlap) tend to penalize methods that generate *novel abductive syntheses*. Since baseline methods (like B3 and B4) construct claims that closely copy or paraphrase literal phrases from the retrieved abstracts, DeBERTa finds high textual overlap, translating to high entailment rates.

Conversely, KG-TABI generates *cross-disciplinary bridging claims* that link topologically isolated communities. Because the claim represents a novel conceptual integration (e.g., combining Android security with disaster logistics), there is no literal or word-level overlap with the topological premise (which only states that the communities are isolated). Consequently, a literal NLI model flags the relationship as neutral, whereas a generative LLM leveraging domain-specific

Table 10. Diagnostic DeBERTa-v3-large NLI Entailment Rates across all evaluated methods.

Method	DeBERTa NLI Entailment Rate
KG-TABI (Ours)	28.3%
B1 (Mulla RAG)	69.0%
B2 (Simple LLM)	36.0%
B3 (GAPMAP Text)	62.4%
B4 (GraphRAG)	63.9%
B5 (LightRAG)	72.0%
B6 (HippoRAG)	59.8%

background knowledge (to process the Warrant) verifies the claim as logically sound. This suggests that strict lexical NLI models are useful for verifying *textual extraction* but are overly conservative for evaluating *conceptual synthesis*.

5.4 Limitations

Several limitations must be acknowledged:

Dataset scale. Our evaluation uses 100 screened papers, producing a knowledge graph of 550 nodes and 425 edges. While sufficient to demonstrate the methodology, larger corpora are needed to assess scalability and the stability of topological metrics. We note that the orphan cluster and decay thresholds (Equations 3–4) may require recalibration for different corpus sizes.

LLM-as-judge bias. The NLI entailment evaluation uses the same LLM backbone that generates the gaps. This introduces potential self-evaluation bias. Future work should employ independent NLI models (e.g., DeBERTa-based NLI) or human judges.

Entity extraction noise. Some extracted entities (e.g., "this study", "current literature") are artifacts of generic scientific prose rather than domain-specific concepts. While the temporal decay analysis correctly flags them as stagnant, the resulting gap statements are less actionable than those derived from orphan clusters. Improved entity type filtering could mitigate this.

Gap validation depth. The current pipeline implements evidence support and NLI checking (validation layers 1–2 from Section 3.6). Full gap closure search (searching for papers that may have already resolved a candidate gap) and importance assessment require additional implementation and are left to future work.

Domain generalizability. The current evaluation uses a cross-domain corpus on "research gap identification." The framework's effectiveness on narrowly-scoped, domain-specific queries (e.g., "microservices security") requires separate evaluation.

6 Conclusion and Future Work

This paper presented KG-TABI, a framework that automates research gap detection by combining Dynamic Knowledge Graphs with Toulmin-Abductive Inference. By identifying isolated communities and stagnant concepts in the literature graph and structuring LLM reasoning through Toulmin's argumentation model, KG-TABI generates evidence-grounded gap candidates that achieve competitive logical entailment (76.1% under LLM verification) while producing structurally novel gaps that are lexically distinct from those identified by text-only and RAG-based baselines.

We formalize a six-type research gap taxonomy and demonstrate that KG-TABI's topological mechanisms naturally detect Coverage Intersection Gaps (via orphan clusters) and Methodological/Evaluation Gaps (via temporal decay). A scalability stress-testing simulation on a synthetically scaled 200-paper graph (5,820 nodes) confirms that our offline pipeline executes in under 27 seconds, and a human expert evaluation confirms high agreement (Kappa = 0.78) for gap candidate verification.

To support transparency and reproducibility, the prompt structures, evaluation datasets, and complete Python source code are published in our public repository at https://github.com/Khoa-Hoa-Technology-Solution-Company/research-paper-gap.

In future work, we plan to: (1) implement the full multi-layer validation pipeline including automated gap closure search via backward and forward citation analysis; (2) scale evaluation to larger corpora (>500 papers) with human expert panels for gold-standard gap annotation; (3) replace LLM-as-NLI-judge with independent entailment models to eliminate self-evaluation bias; and (4) extend the gap taxonomy to support automated classification of detected gaps into the six defined types.

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